Yahya Masri

(703) 420-0485 | yahya.masri@yahoo.com | linkedin.com/in/yahya-masri | yahyamasri.com | github.com/yahya3867

Education

Bachelor of Science in Computer Science — Expected May 2027
George Mason University — *Fairfax, VA*

- GPA: 3.92 / 4.00 | Dean's List (2x)
- Relevant Courses: Data Structures & Algorithms, Software Engineering, Object-Oriented Programming, Computer Systems and Programming, Concurrent & Distributed Systems, Natural Language Processing

Experience

SAP NS2 — May 2026 – Present
Software Engineer Intern — *Herndon, VA*

- Built an AI usage analytics agent to transform **400K+ internal AI interactions** across **847 users** into leadership analytics, enriching **AWS OpenSearch** data with business-line, use-case, token, and time-savings metadata.
- Increased enrichment throughput by **5.6x**, from **25 to 140 records/min**, by replacing page-synchronous processing with **continuous work scheduling** across **12 concurrent workers** and horizontally scaled instances.
- Secured a **CUI-authorized** internal AI platform with role-scoped access controls below the LLM, using **adversarial red-team testing** to uncover and remediate identity-leakage paths.
- Won **1st place overall among 300 participants** by building an agentic workspace using live **Microsoft Graph API** data to surface meetings, priorities, and follow-ups.

Lillup — Jan 2026 – Mar 2026
ML Engineer Intern — *Remote*

- Achieved **90% source-grounded responses** at **2s query latency** by building a local CPU-based RAG pipeline with **ChromaDB**, **llama-cpp-python**, and GBNF-constrained JSON output for an on-device learning assistant.
- Cut time-to-first-audio from **7s to 500ms** (14x) on CPU by streaming LLM output sentence-by-sentence into TTS and switching from a 3B Llama model to **Qwen3-0.6B** for faster on-device inference.

NSF Spatiotemporal Innovation Center, GMU — Feb 2024 – Present
Research Assistant — *Fairfax, VA*

- Built an LLM-enabled digital twin for a **600+ machine data center**, integrating Prometheus/Loki telemetry, PostgreSQL, and **NVIDIA Omniverse** through **19 MCP tools** for natural-language diagnostics and 3D control.
- Published **5 peer-reviewed publications, including 3 as first author**, on LLM/SLM evaluation, RAG, digital twins, and AI-driven conflict analysis.
- Built a real-time conflict-intelligence system that continuously ingests global news, uses **LLMs** to extract and validate grounded incidents, and streams geocoded events to a **Next.js/MapLibre** dashboard.

Technical Skills

Languages: Python, Java, C, SQL, Rust
AI/ML & Frameworks: PyTorch, FastAPI, LangChain, MCP, RAG, HuggingFace, ChromaDB, FAISS, scikit-learn
Cloud & Infrastructure: AWS, Docker, Kubernetes, Linux, OpenSearch, PostgreSQL, MongoDB, Prometheus, Grafana, Git

Projects

Autonomous FPV Drone Racing System | *Anduril — AI Grand Prix* | GitHub — Jul 2026

- Built a closed-loop autonomous drone-racing system using **FPV computer vision**, a **38D state representation**, and **PPO reinforcement learning**; qualified for **Round 2** of the Anduril AI Grand Prix.
- Engineered a **Gymnasium/PPO** training pipeline with reward shaping, curriculum learning, deterministic evaluation, and live reward/loss monitoring for autonomous gate navigation.

Autonomous Solana Trading Agent | *Winner — Best Use of Solana Track* | GitHub — Feb 2026

- Built a server-side **risk engine** for an autonomous Solana trading agent, enforcing order-size, exposure, and confidence limits plus an **instant kill switch** before any paper-trade execution.
- Deployed the agentic trading stack on **AWS EC2** with **Bedrock** and **RDS PostgreSQL**, enabling end-to-end autonomous trade cycles from strategy generation through risk approval, execution, and persistence.